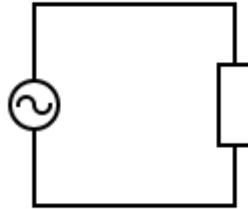
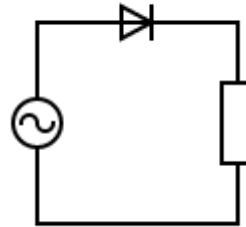


27	A circuit with a sinusoidal alternating power supply is rectified with a diode as shown.
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[Turn over



before rectification



after rectification

The mean power of the circuit before rectification is P .

What is the peak and mean power of the circuit after rectification?

		peak power	mean power	
	A	$0.5P$	$0.5P$	
	B	P	$0.5P$	
	C	$2P$	$0.5P$	
	D	$2P$	P	

Ans: C

Half-wave rectification does not change peak power but halves mean power.

Peak power = twice mean power for sinusoidal alternating power supply
 $= 2P$